

## CONTACT INFORMATION

University of Kansas (KU)  
Department of Physics and Astronomy  
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## PROFESSIONAL EXPERIENCE

**Graduate Research Assistant**  
University of Kansas  
Supervisor: Dr. Ian J. M. Crossfield,  
Associate Professor, KU

January 2025- Present

**Graduate Teaching Assistant**  
University of Kansas  
Demonstrated Labs and Grading  
Supervisor: Dr. Jennifer A. Delgado,  
Associate Teaching Professor, KU

August 2024 – December 2024

**Senior Lecturer** in Physics  
Northern University Bangladesh (NUB)

August 2023 – to June 2024

**Lecturer** in Physics  
Northern University Bangladesh (NUB)

September 2019 – June 2023

## EDUCATION

**M. S. in Physics**

January 2018 - May 2019

Position: 5<sup>th</sup> (Out of 75)

Thesis Title: Study of SMES device and SMES-PCC computer program development for the analysis of superconducting coil design

Supervisor: Dr. Md. Rafiqul Islam, Professor, University of Chittagong

**B. Sc. in Physics**

January 2013 - January 2018

Department of Physics , University of Chittagong, Bangladesh

Position: 4<sup>th</sup> (Out of 92)

## RESEARCH INTERESTS

- Astronomical Data Analysis
- Exoplanet Detection
- Planetary Atmosphere
- Biosignatures, life in Space
- Space Mission design
- Space Radiation
- Radiation Shielding

## PUBLICATIONS

**Citations :** 145 (h-index: 6) \*Source: Researchgate

## PEER REVIEWED JOURNAL PAPERS

1. **Al Zaman, MD. A.,** & Kunja, L. A. (2023). Effectiveness of radiation shields constructed from Martian regolith and different polymers for human habitat on Mars using MULASSIS/GEANT4 and OLTARIS. AIP Advances, 13(8).
2. **Al Zaman, MD. A.,** & Monira, N. J. (2023). Shielding effectiveness of different polymers and low-density hydrides in a combined radiation shield for crewed interplanetary space missions. Radiation Physics and Chemistry, 205, 110706.
3. **Al Zaman, MD. A.,** Roy, S., Sarker, S., Azad, M. M. U., Akib, A. A., & Haque, M. M. (2023). Comparative Analysis of the Effectiveness of Natural Polymers and Conventional Space Radiation Shielding Polymers in Spacecraft for Prolonged Space Expeditions. Materials Research Express.
4. **Al Zaman, MD. A.,** Islam, M. R., Maruf, H. M. A. R., & Nizam, Q. M. R. (2022). Effectiveness of Kevlar and water-soaked hygienic wipes in a combined radiation

shield for manned long termed space missions. Radiation Physics and Chemistry, 110483.

5. **Al Zaman, MD. A.**, Nizam, Q.M.R. (2022) Study on Shielding Effectiveness of a Combined Radiation Shield for Manned Long Termed Interplanetary Expeditions. Journal of Space Safety Engineering <https://doi.org/10.1016/j.jsse.2021.12.003>
6. **Al Zaman, MD. A.**, Maruf, H. M. A. R., Islam, M. R. (2021). Study on superconducting magnetic shield for the manned long termed space voyages. The Egyptian Journal of Remote Sensing and Space Science, 24(2), 203-210
7. **Al Zaman, MD. A.**, Islam, M., & Maruf, H. (2020). Study on Conceptual Designs of Superconducting Coil for Energy Storage in SMES. East European Journal of Physics, (1), 111-120.
8. Islam, M. R., **Al Zaman, MD. A.** (2020) SMES-PCC computer program development for studying and designing the superconducting magnets for energy storage purpose. Solid State Science and Technology, v. 27, n. 1 & 2, 40 – 50.

## CONFERENCE PRESENTATIONS

1. **Al Zaman, MD. A.**, Gupta, Sanchita Das, Maruf, H. M. A. R, Gupta, Somoy Das (2025) Estimating Low-Density Hydride Effectiveness in Spacesuit Radiation Shielding: A Comparative Computational Approach, KU Physics & Astronomy Locally OrganiZed Assembly (PALOOZA'25) (<https://indico.ku.edu/event/480/timetable/>)
2. **Al Zaman, MD. A.**, Kunja, L. A. (2024) Radiation Shielding Strategies for Human Habitats on Mars: Exploring the Effectiveness of Shields Constructed with Martian Regolith and Hydrogenous Polymers Using Geant4 and OLTARIS. 'Mid-American Regional Astrophysics Conference (MARAC)', December 5-6, 2024
3. **Al Zaman, MD. A.**, Islam, M. R., Maruf, H. M. A. R.(2023) Star Formation Rate Profiles in Spiral Galaxies at  $z < 0.03$ . 5th International Conference on "Physics for Sustainable Development and Technology (ICPSDT-2023), September 7-8, 2023
4. Kunja, L. A., **Al Zaman, MD. A.** (2023) Effectiveness of radiation shields constructed from Martian regolith and different polymers for human habitat on Mars using MULASSIS/GEANT4 and OLTARIS. Fall 2023 Academic Conference and 41<sup>st</sup> Annual General Meeting, Korean Space Science Society (25 - 27<sup>th</sup> October 2023)
5. **Al Zaman, MD. A.**, Islam, M. R., Maruf, H. M. A. R., Nizam, QMR (2022) Study on Effectiveness of Polyimide and Polyester Polymer for Radiation Shielding in a Spacecraft for Long Termed Space Mission. International Conference on Physics 2022, May 19-21, 2022, Organized by Bangladesh Physical Society at Atomic Energy Center Dhaka.
6. **Al Zaman, MD. A.**, Maruf, H. M. A. R., Islam, M. R. (2020) 'Study on Superconducting Magnetic Shield for the Manned Space Expeditions.' Int. conference on Physics (2020) at Atomic energy Center, Dhaka, Bangladesh (5-7th March '20), Bangladesh Physical Society.
7. **Al Zaman, MD. A.**, Islam, M. R.(2019) 'Study of Design Optimization of Superconducting Solenoid and Toroid Coil for the Energy Storage in SMES'. National conference on Physics '19 (7-8<sup>th</sup> February 2019), Bangladesh Physical Society

## CONFERENCE POSTERS

1. **Al Zaman, MD. A.**, Crossfield, Ian J. M., Brande, Jonathan (2025) A New Eye in Space: Exploring the Potential of a Next-Generation Space Telescope for Exoplanet Science Cases. KU Physics & Astronomy LOcally OrganiZed Assembly (PALOOZA'25, 26<sup>th</sup> April)  
<https://indico.ku.edu/event/480/sessions/613/#20250426>
2. **Al Zaman, MD. A.**, Islam, M. R., Maruf, H. M. A. R., Nizam, QMR (2022) Study on the effectiveness of Kevlar and Polyethylene for radiation shielding in a spacecraft for long term space missions. *4th International Conference on "Physics for Sustainable Development and Technology (ICPSDT-2022), Jan 22-23, 2022*
3. **Al Zaman, MD. A.**, Maruf, H. M. A. R., Islam, M. R. (2021) 'Study on Shielding Effectiveness of a Conceptual Active Radiation Shield Constructed with Multi-Wall Carbon Nanotube Added YBCO-123 Superconducting Material' *National Conference on Physics (6-7 August '21), Bangladesh Physical Society.*
4. **Al Zaman, MD. A.** et.al. (2019) Space applications of High Tc Superconductivity and Superconducting Magnetic shield for manned space missions. *3rd International Conference on "Physics for Sustainable Development & Technology (ICPSDT-2019) (18 -19th December '19).*
5. **Al Zaman, MD. A.**, Ahmed, S., Monira, N. J. (2018) An overview of Superconducting Magnetic Energy Storage (SMES) and Its Applications. *International Conference on Nanotechnology and Condensed Matter Physics 2018 (ICNCMP 2018)*

## ASTRONOMICAL OBSERVING EXPERIENCE

### W. M. Keck Observatory

Instrument: KPF 1

Mode: Remote

9 Half Nights

## AWARDS & HONORS

### Best Faculty Member for Contribution to the Development and Growth

Department of Textile Engineering,  
Northern University Bangladesh  
On the Occasion of National Textile Day'23

December 12, 2023

### Best presenter (Oral Presentation)

- 5<sup>th</sup> International Conference on "Physics for Sustainable Development and Technology (ICPSDT-2023)

September 8, 2023

### 2 Best Poster Awards

- 4<sup>th</sup> International Conference on "Physics for Sustainable Development and Technology (ICPSDT-2022)
- National Conference on Physics 2021

January 22 - 23, 2022

August 6 - 7, 2021

### National Finalist

EATL-Prothom Alo Apps Challenge 2016  
Lead Sponsor: World Bank  
For: Emergency Bangladesh (an Android Utility Application)

11<sup>th</sup> February 2017

## TEACHING (Undergraduate)

PHSX 216: General Physics I Laboratory LAB at KU

Fall 2024

PHY 1305 & PHY 1306

(Physics II & Physics Lab: Electricity and Magnetism, Heat & Thermodynamics)

(40 Students Per Section)

Fall 2019 – Spring 2024

EEE 3105 & EEE 3106

(Basic Electrical and Electronics Engineering & Lab)

(40 Students Per Section)

Spring'21, Summer'21

PHY 1109 & PHY 1110

(Physics I & Physics Lab: Properties of Matter and Optics )

Fall 2019 – Fall 2023

## COMPUTER SKILLS

### Programming Languages

C, C++, Python

### Softwares

TOPCAT, DS9

MATLAB, SPENVIS (GEANT4), OLTARIS

### Operating Systems

Linux (Ubuntu), Windows

## RESPONSIBILITIES

1. **Judge (Poster)**, Undergrad Research Symposium, University of Kansas, 25<sup>th</sup> April, 2025
2. **REU Application Reviewer Panel** (Summer 2025), Department of Physics and Astronomy, University of Kansas
3. **Convener**, Seminar on Impact of Artificial Intelligence on Education and Industries, 31<sup>st</sup> July 2023
4. **Co-Convener**, Celebration of National Textile Day, 12<sup>th</sup> December 2023
5. **Convener**, Seminar on Contemporary Research in Department of Textile Engineering, NUB, 12<sup>th</sup> November, 2022
6. **Convener**, Workshop on MATLAB, 31<sup>st</sup> July 2022
7. **Convener**: Promotion, Digital Activities and Publications, Department of Textile Engineering, NUB (From September 2023 to June 2024)
8. **Co - Convener**, Celebration of National Textile Day, 4<sup>th</sup> December 2022
9. **Technical Member**, Journal of Natural Science and Textile Technologies (JNSTT), an online journal published by Department of Textile Engineering, Northern University Bangladesh <https://jnstt-nubtex.com/>
10. **Technical Member**, Textile Newsletter, a Semester wise online Newsletter, published by Department of textile Engineering, Northern University Bangladesh <https://textile.nub.ac.bd/page/54/Newsletter>
11. **Advisor**, NUB Textile Research and Publications Club (From July 2021 – January 2023)
12. **Member**, Outcome Based Education Curriculum Formulation Committee, Department of Textile Engineering
13. **Member** (Documentation), Central Admission Committee, Northern University Bangladesh (January 2022- February 2024)
14. **Co-Convener** (January 2021- August 2022), Documentation Committee, Department of Textile Engineering, NUB
15. **Former Executive Committee Member**, Chittagong University Physics Society

## OUTREACH

- Worked as a **GRE and TOEFL Instructor** (Around 10 months) in a Higher Study Consultancy Center Called Nextop-USA (<https://nextopusa.com/>)
- **Feature writer** (The Business Standard, Roar media, Bigganjatra)  
Wrote around 30 articles covering many topics on Physics, Astronomy, Education, Technology, Book review.  
**The Business Standard Article:** <https://www.tbsnews.net/thoughts/judging-quantity-how-well-bangladesh-faring-research-505990>  
  
**Roar Media Writer Profile:** <https://archive.roar.media/author/alzaman>  
**BigganJatra** (Science-Sailing) **Writer Profile:** <https://bigganjatra.org/writer/proyash-zaman/>
- Write about research tools, literature review, predatory journals/practices and others in personal blog.  
Link: <https://cuprw18.wixsite.com/home/research-blog>
- Volunteered at Jamal Nazrul Islam Science Fair (2016) (Demonstrated some basic Physics Experiments to the students in elementary level)
- Volunteered at Bangladesh Physics Olympiad-Chittagong Chapter (2016) (As an examiner. Evaluated answers scripts)